A Robust and Reliable Machine Learning Approach for Carbon Star Identification in LAMOST DR9 and Gaia DR3

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1. Abstract

Carbon stars are important distance indicators and tracers of stellar evolution and Galactic structure. However, their automatic identification in large spectroscopic surveys remains challenging due to the limited number of confirmed carbon stars available for model development and validation. We present a robust and computationally efficient framework that combines LAMOST DR9 spectroscopy with Gaia DR3 photometry for carbon-star detection. A Gradient Boosting Machine (GBM) classifier optimized through a modified Monte Carlo Cross-Validation framework was applied to approximately 4.8 million spectra. The proposed methodology achieved over 97% performance in the identification of known Galactic carbon stars and identified 6372 Galactic carbon-star candidates, including 5665 previously unreported sources. Photometric and spectroscopic validation analyses support the astrophysical plausibility of the identified population, demonstrating the potential of machine-learning techniques for large-scale carbon-star searches in modern astronomical surveys.

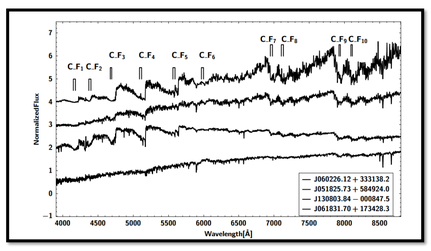
2. Introduction

Carbon stars are evolved stars with carbon-rich atmospheres that provide key information about stellar evolution and Galactic chemical enrichment. Despite their astrophysical relevance, they represent a very small fraction of sources in modern surveys, making their identification challenging. We developed a machine-learning framework [1] that combines LAMOST DR9 [2,3] spectroscopy and Gaia DR3 [4] photometry to efficiently identify Galactic carbon-star candidates.

3. Aim



LAMOST Spectroscopy



Characteristic spectral signatures of carbon stars [1].

Gaia Photometry & Astrometry

$$BP\text{-}RP$$

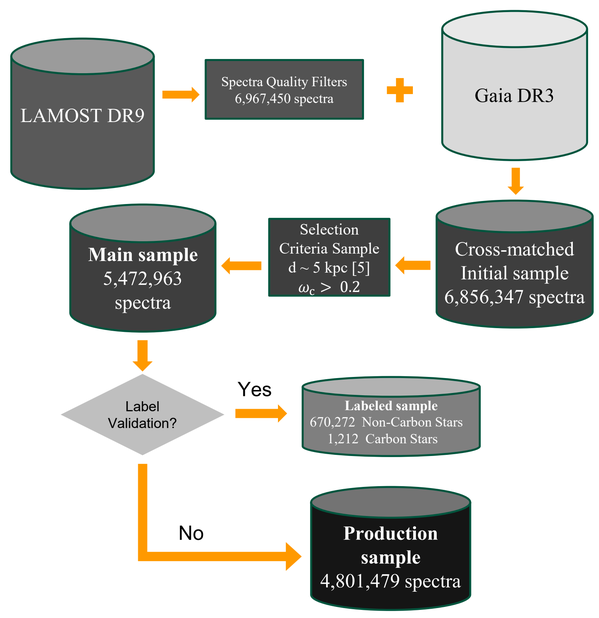
$$M_{RP} = RP - 5\log\left({}^{1000}/_{\omega_c}\right) + 5$$

Automated Identification of Galactic Carbon Stars

Adapted from [1].

4. Method

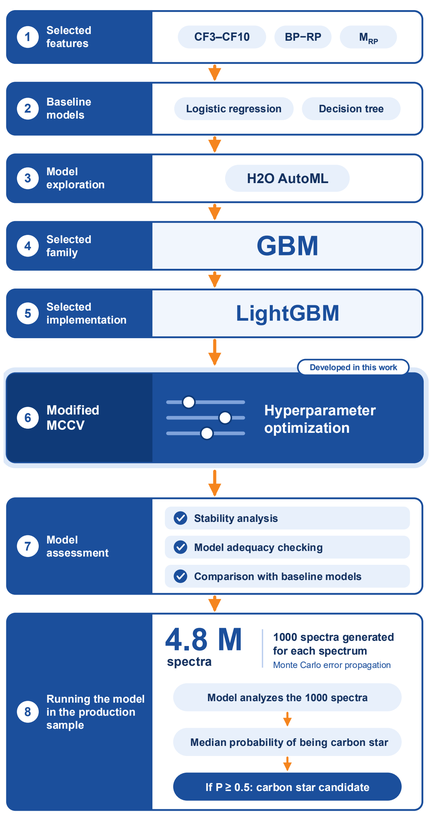
Sample Construction Workflow



Construction of the labeled and production samples from the cross-matched LAMOST DR9 and Gaia DR3 catalogs after quality filtering, sample selection, and label validation. The labeled sample was used for model training and cross-validation, whereas the production sample was used to run the proposed model [1].

THE SELECTED 10 FEATURES			
C_2 Bands		CN Bands	
Feature	λ	Feature	λ
CF3	4680 Å	CF7	6960 Å
CF4	5100 Å	CF8	7116 Å
CF5	5575 Å	CF9	7930 Å
CF6	5980 Å	CF10	8100 Å
Gaia DR3: BP-RP y M_{RP}			

Selected features used for model training. CF3-CF10 correspond to spectral indices associated with prominent C_2 and CN molecular bands, complemented by the Gaia DR3 features $BP-RP$ and M_{RP} [1,6,7].



The selected Gradient Boosting Machine (GBM) classifier was implemented using LightGBM [8] and optimized through a modified Monte Carlo Cross-Validation (MCCV) framework [1]. Hyperparameter configurations were ranked according to predictive performance and statistical convergence, assessed from the cumulative F1-macro behavior across iterations. Production-sample classifications were obtained from Monte Carlo realizations generated from the observational uncertainties of the input features.

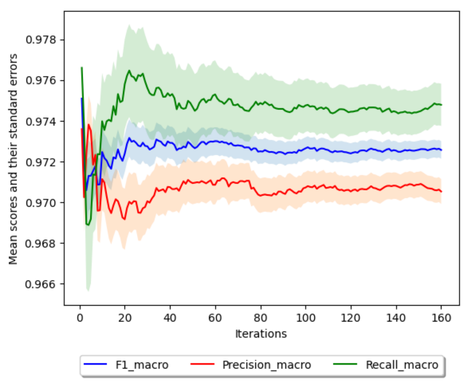
5. Results

GBM Model Assessment

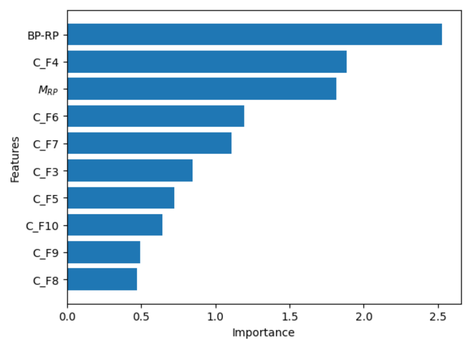


Shells and arcs around carbon star CW Leonis.

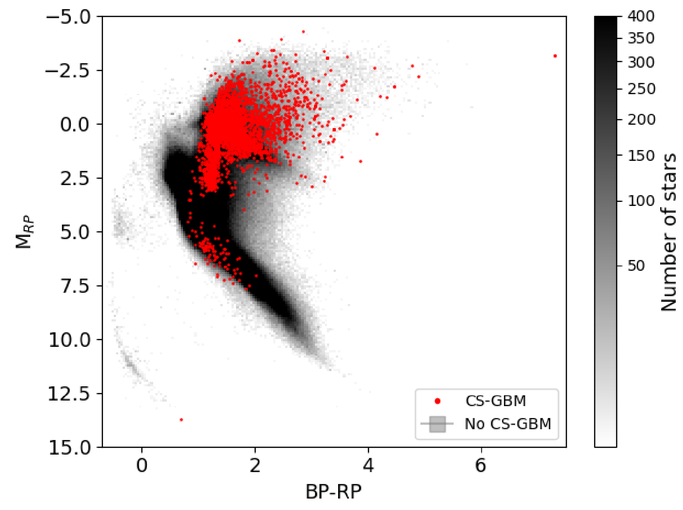
Astronomy Picture of the Day (APOD) 2023 July 17, Taken from https://science.nasa.gov/image-article/apod-2023-july-17-shells-and-arcs-around-star-cw-leonis/



Evolution of the cumulative mean F1-macro, precision, and recall scores for the proposed GBM classifier during the modified MCCV procedure. The optimal configuration was selected through convergence of the cumulative mean F1-macro score and its associated standard deviation across iterations. Shaded regions represent the evolving standard deviations.

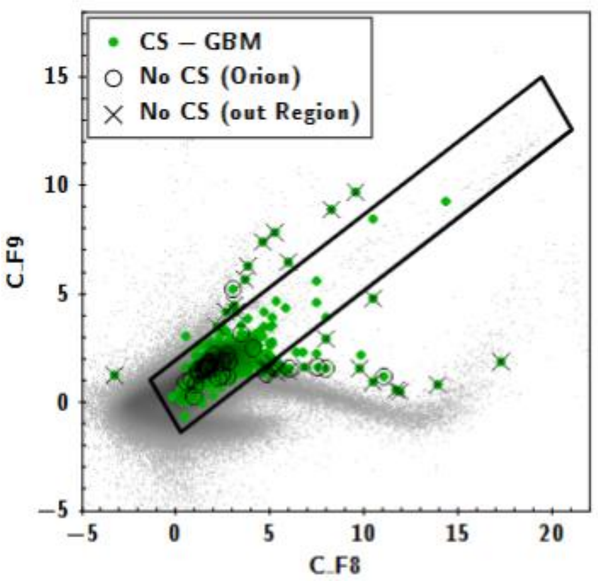


SHAP summary plot showing the contribution of the selected features to the predictions of the proposed GBM model. Higher absolute SHAP values indicate greater influence on carbon-star classification [1].



Color-magnitude distribution of the 6372 carbon-star candidates (CS-GBM) detected by the proposed GBM classifier and reported in [1]. The candidate population spans the region commonly occupied by carbon-rich giant stars.

Candidate Validation in the Orion Region



Distribution of the 6372 CS-GBM detected by the proposed GBM classifier in the CF8-CF9 feature space. The rectangle denotes the region occupied by known carbon stars (CS), allowing a spectroscopic validation of the candidate population.[1]

6. Conclusions

- 6372 Galactic carbon-star candidates were identified in the production sample, including 5665 previously unreported sources.
- The proposed GBM + modified MCCV framework achieved robust carbon-star detection (F1-macro = 0.973).
- Visual inspection of Orion-region candidates confirmed carbon-rich spectral signatures in >94% of the evaluated detections.
- Production-scale inference on 4.8 million spectra with 1000 Monte Carlo realizations per source was completed in approximately 9 h on a personal workstation

7. References

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Full paper

Poster